Tutorial 11: Evaluating Insurance and Annuity Products with Excel

(TEACHER'S VERSION WITH EXPANDED SOLUTIONS)

Theme: Life Tables, Expected Present Value, Withdrawal Decisions, Marriage Insurance and Single Annuity

Programme: Nurturing Future Risk Managers and Actuaries: An AI-powered Journey for Gifted Secondary Students

A. From Tutorial 10 to Tutorial 11: Why This Tutorial Exists

Synergy & Learning Flow

Connection with Tutorial 10: In Tutorial 10, students learned how to build a reliable Excel model using clear inputs, formulas, outputs and checks. They also learned expected value, present value, expected present value, lookup formulas, sensitivity thinking and AI-supported formula debugging.

Purpose of Tutorial 11: In this tutorial, students apply those Excel modelling habits to actuarial-style product evaluation. Instead of using small artificial examples, students now work with life table information, insurance cash flows, withdrawal values, bonus values, and demographic data related to marital status.

Teacher framing: This is not a professional actuarial examination. It is a guided modelling experience. The main question is not only "Can students calculate?" The deeper question is "Can students explain what the spreadsheet is doing and why the result makes financial sense?"

Key Definition / Result

A future risk manager or actuary should not only ask:

What is the answer?

A future risk manager or actuary should also ask:

What assumptions are used? What probabilities are used? When is the money paid? How is the future money discounted? Can the result be checked?

In this tutorial, the key modelling pattern is:

$$\text{Expected Present Value} = \text{Cash Flow} \times \text{Probability} \times \text{Discount Factor}.$$

This single idea will appear repeatedly in insurance, withdrawal value, marriage insurance and single annuity calculations.

Suggested AI Prompt: Preparing for Tutorial 11 with Gemini ⋆ **Core**

"Hi Gemini. I have just learned Excel spreadsheet modelling, expected value, present value and expected present value. I am now going to study an insurance valuation tutorial using life tables and cash flows. Please explain, in beginner-friendly language, how amount, probability and timing work together in an actuarial spreadsheet model."

B. Reading the Insurance Workbook: Variables and Meaning

Synergy & Learning Flow

Why this section matters: Before calculating anything, students must understand what each column in the insurance workbook means. A common student mistake is to jump directly into formulas without reading the business meaning of the variables. In actuarial modelling, every column should represent a financial or probabilistic idea.

Investigation / Excel Mission: Understand the Insurance Summary Table ⋆ **Core**

The first workbook contains a summary table of a life insurance product. Students should identify the role of each variable before writing formulas.

Variable in Excel file	Meaning
Term	Policy year. This tells us which year of the insurance contract we are considering.
Age	Age of the insured person at that policy year.
Face Value	Cash that a beneficiary receives upon the death of the insured.
Face Value plus Bonus	Face value after adding policy bonus.
Premium	Payment made by the policyholder to the insurance company.
Withdrawal	Amount received if the insured chooses to cash in by abandoning the insurance plan.
Withdrawal plus Bonus	Withdrawal amount after adding bonus.
Discount factor	Factor used to convert future money back to today's value.
Death Rate	Probability that death occurs during the relevant age interval.
Survival Rate	Probability that the insured survives from the starting age to the relevant point.
Expected PV	Probability-weighted and discounted value of a future cash flow.
PnL	Profit or loss from the viewpoint of comparing premium value and benefit value.

Teacher Note: Ask students to mark columns as Input, Probability, Discounting, Expected Value, Output or Check. This directly continues Tutorial 10's spreadsheet structure mindset.

Key Definition / Result

Face Value: Cash that a beneficiary receives upon the death of the insured.

Bonus: The insurance company may invest the collected premiums. If the investment experience is favourable, additional value may be added to the policy as a bonus.

Withdrawal: The insured may choose to cash in by abandoning the insurance plan. This means the insured stops the policy and receives a withdrawal value instead of continuing the contract.

Important modelling idea: A benefit may look large in future dollars, but actuarial valuation must consider whether the benefit is likely to be paid and how far in the future it is paid.

Reflection Question: 1

Reading Before Calculating Why is it dangerous to use a spreadsheet formula before understanding what the columns mean?

Suggested AI Prompt: Understanding Insurance Variables ⋆ **Core**

"Hi Gemini. I am reading an Excel workbook for a life insurance valuation exercise. The columns include Term, Age, Face Value, Face Value plus Bonus, Premium, Withdrawal, Withdrawal plus Bonus, Death Rate, Survival Rate, Discount Factor and Expected PV. Please explain each column in simple language and tell me whether each column is an input, a probability, a calculation or an output."

C. Life Table Thinking: Survival and Death Probabilities

Synergy & Learning Flow

Why this section matters: The life table is the probability engine of the insurance model. Students do not need professional life contingency notation, but they should understand the basic row-by-row logic: out of many people alive at one age, some die before the next age, and the remaining people survive.

Key Definition / Result

A life table usually contains the following ideas:

- **Age:** the age being considered;
- **Survivors:** the number of people still alive at that age;
- **Deaths:** the number of people who die before reaching the next age;
- **Death rate:** the chance that a person dies during that age interval;
- **Survival rate:** the chance that a person survives from the starting age to the relevant age.

In beginner-friendly language:

$$\text{Death rate} = \frac{\text{Number of deaths during the interval}}{\text{Number of people alive at the start of the interval}}.$$

The survival rate tells us how likely the policyholder is to still be around when a future cash flow is considered.

Investigation / Excel Mission: Use the Life Table in Excel ⋆ Core

Task. Using the life table and insurance summary attached in the Excel file, answer the following conceptual questions before calculating final values.

1. Which column tells us the age of the insured?
2. Which column tells us the chance of death during an age interval?
3. Which column tells us the chance that the insured has survived from the starting point?
4. Why do we need both death rate and survival rate when evaluating a life insurance product?
5. Why is it not enough to look only at the face value?

Teacher Note: Students should explain in words. A good answer is that face value tells us the amount paid if the event happens, while the death and survival probabilities tell us how likely the event and timing are.

Suggested AI Prompt: Learning Life Tables with Gemini ⋆ Core

"Using Google Gemini as my learning assistant, show me how to use a life table in an Excel file to calculate the probability of survival and death for different age groups. Explain what each column and row in the life table means and how the death rate and survival rate are derived. Please explain in a way suitable for a secondary school student who is just beginning actuarial modelling."

Common Mistake / Debugging Warning

Students may confuse the death rate during one year with the total probability of surviving many years. A one-year death rate is a local probability for one interval. A survival rate from the starting age is a cumulative probability across time.

D. Discount Factors and Present Value in the Insurance Workbook

Synergy & Learning Flow

Why this section matters: Tutorial 10 introduced present value and expected present value. Tutorial 11 now applies these ideas to real insurance cash flows. Students should see why future money cannot simply be compared with today's money without discounting.

Key Definition / Result

If the annual interest rate is r, the discount factor for a cash flow paid at time t is:

$$\text{Discount Factor} = \frac{1}{(1+r)^t}.$$

Present value is:

$$\text{PV} = \text{Future Cash Flow} \times \text{Discount Factor}.$$

Expected present value is:

$$\text{EPV} = \text{Future Cash Flow} \times \text{Probability} \times \text{Discount Factor}.$$

In an insurance model, the probability may involve survival to reach a certain age and death during a particular age interval.

Investigation / Excel Mission: Using Excel's PV Thinking ⋆ Core

Task. Assume the interest rate is 3%. Students should identify where the discount factor appears in the workbook and explain why discounting is needed.

Suggested Excel formula for a manual discount factor, if the interest rate is stored in a named cell called `Interest_Rate` and the term is in `A2`:

```
=1/(1+Interest_Rate)^A2
```

Discussion questions:

1. What happens to the discount factor when the term becomes larger?
2. Why is a cash flow paid far in the future worth less today?
3. Why does an actuary need both probability and discounting?

Teacher Note: Students should say that the discount factor decreases as time increases. The expected present value therefore becomes smaller when payment is less likely or further away.

Suggested AI Prompt: Understanding the PV Function ⋆ **Core**

"Using Google Gemini as my learning assistant, show me how to use the PV function in Microsoft Excel to calculate the present value of a future cash flow. Explain the arguments of the PV function and how they affect the result. Then compare this with manually using the formula Future Amount divided by (1 plus interest rate) to the power of time."

Suggested AI Prompt: Explaining Discount Factors ⋆ **Core**

"Gemini, I am confused about discount factors in an insurance spreadsheet. Please explain why a future benefit of $100,000 is not valued as $100,000 today. Use a simple example with an interest rate of 3% and explain why the discount factor decreases over time."

E. Insurance Evaluation Task 1: No Withdrawal Scenario

Synergy & Learning Flow

Purpose of this section: Students now evaluate the insurance product assuming that the insured will not withdraw from the plan. This helps them focus on premium, face value, bonus and expected present value without the extra complication of withdrawal timing.

Investigation / Excel Mission: Evaluate the Insurance Without Withdrawal ⋆⋆
Intermediate

Using the fact that the interest rate is 3%, and using the life table and insurance summary in the Excel file, answer the following questions.

Given that the insured will not withdraw from this insurance plan:

1. What is the expected PV of your premium?
2. What is the expected PV of the face value?

3. What is the expected PV of the face value plus bonus?

Expanded solution:

$$\text{Expected PV of Premium} \approx \mathbf{21002}.$$

$$\text{Expected PV of Face Value} \approx \mathbf{12309}.$$

$$\text{Expected PV of Face Value plus Bonus} \approx \mathbf{50385}.$$

Interpretation: The expected PV of the premium is the probability-weighted and discounted value of premium payments. It is not just the nominal sum of all possible premiums. The expected PV of face value is smaller than the raw face value because the benefit is paid only if death occurs at certain times and because future payments are discounted. The expected PV of face value plus bonus is much larger because the bonus-adjusted benefit can grow substantially over time.
Teacher Note: Ask students to locate these cumulative expected present value columns in the spreadsheet. They should explain whether they are reading a row value or a cumulative total.

Suggested AI Prompt: Checking the Insurance Evaluation Formula ⋆⋆
Intermediate

"Gemini, I am evaluating a life insurance product in Excel. The workbook has columns for premium, face value, face value plus bonus, survival probability, death rate and discount factor. Please explain how to calculate the expected present value of premium and expected present value of benefit. Do not just give me the answer. Explain what each multiplication represents."

Reflection Question: 1

No Withdrawal Scenario Why is the expected PV of the face value not equal to the face value printed in the policy?

F. Insurance Evaluation Task 2: Withdrawal at the End of Age 65

Synergy & Learning Flow

Purpose of this section: Withdrawal introduces a new type of cash flow. Students must distinguish between continuing the policy and cashing in the plan at a selected age. This section helps students understand that actuarial value depends on behaviour as well as mortality and discounting.

Investigation / Excel Mission: Evaluate the Insurance with Withdrawal at Age 65

⋆⋆ **Intermediate**

Using the same interest rate of 3%, answer the following questions.

Given that the insured will withdraw at the end of age 65:

1. What is the expected PV of the face value?
2. What is the expected PV of the face value plus bonus?
3. What is the expected PV of the withdrawal?

Expanded solution:

$$\text{Expected PV of Face Value} \approx \mathbf{2981}.$$

$$\text{Expected PV of Face Value plus Bonus} \approx \mathbf{5308}.$$

$$\text{Expected PV of Withdrawal} \approx \mathbf{7262}.$$

Interpretation: If the insured withdraws at the end of age 65, the valuation no longer considers all later insurance years in the same way. Future death benefits after withdrawal are no longer relevant under that scenario. Instead, the model includes the expected present value of the withdrawal amount, adjusted for the probability that the insured reaches that withdrawal age and discounted back to today. **Teacher Note: Emphasise the scenario wording. Students should not mix the no-withdrawal scenario and the age-65 withdrawal scenario. Each scenario has its own cash-flow path.**

Suggested AI Prompt: Explaining Withdrawal Value ⋆⋆ **Intermediate**

"Gemini, in my insurance spreadsheet, the insured may either continue the policy or withdraw at a specific age. Please explain how withdrawal changes the expected present value calculation. Why should I stop considering later death benefits after the withdrawal point? Explain using simple cash-flow logic."

Common Mistake / Debugging Warning

A common mistake is to add the withdrawal value and all future death benefits after the withdrawal date. This double-counts benefits. Once the insured withdraws and cashes in the plan, the later insurance benefits should not be treated as if the policy were still continuing.

G. Insurance Evaluation Task 3: Best and Worst Withdrawal Time

Synergy & Learning Flow

Purpose of this section: This section asks students to compare withdrawal timing. The goal is not only to find maximum or minimum numbers in Excel, but also to interpret why timing matters and why the conclusion may change when policy bonus is included.

Investigation / Excel Mission: Find the Best and Worst Withdrawal Time ⋆⋆ **Intermediate**

Task. Using the PnL columns in the insurance workbook, answer the following question.

Question: What is the best and worst time for the insured to withdraw from the scheme?

No bonus:

- If the insured withdraws at term 1, the expected loss is about **992**. This is the best result in the no-bonus withdrawal comparison because it is the smallest expected loss.
- If the insured withdraws at term 45, the expected loss is about **10830**. This is the worst result in the no-bonus withdrawal comparison because it is the largest expected loss.

With bonus:

- If the insured withdraws at term 76, the expected gain is about **29415**. This is the best result in the bonus-adjusted withdrawal comparison under this workbook interpretation.
- If the insured withdraws at term 12, the expected loss is about **6461**. This is the worst result in the bonus-adjusted withdrawal comparison under this workbook interpretation.

Important correction for teacher use: The value 10830 belongs to the worst no-bonus case. It should not be used as the term-12 loss in the bonus-adjusted case. In the bonus-adjusted column, the term-12 loss is approximately 6461.
Teacher Note: This is an excellent chance to teach students that a spreadsheet may contain multiple PnL columns. They must read the correct column, not only search for a familiar number.

Suggested AI Prompt: Finding Maximum and Minimum PnL in Excel ⋆⋆ **Intermediate**

"Gemini, I have an Excel table with withdrawal terms and two PnL columns: one without bonus and one with bonus. Please show me how to identify the best and worst withdrawal timing using MAX, MIN, XLOOKUP or FILTER. Also remind me how to avoid mixing up the no-bonus and with-bonus columns."

Suggested AI Prompt: Interpreting Withdrawal Timing ⋆⋆ **Intermediate**

"Gemini, I found that the best withdrawal time changes when bonus is included. Please help me explain why policy bonus can change the best timing decision. Use simple language about future benefits, discounting and accumulated bonus value."

Reflection Question: 1

Best and Worst Timing Why can the best withdrawal time be different when bonus is included?

H. AI-supported Insurance Evaluation Protocol

Synergy & Learning Flow

How students should use Gemini in this tutorial: Students should use Gemini to understand formulas, review spreadsheet structures, debug cell references and improve interpretation. They should not use Gemini to replace their own checking. Every AI-generated formula must be checked against the workbook layout.

Key Definition / Result

A good AI-assisted insurance evaluation workflow:

1. Read the variable names and understand the cash-flow meaning.
2. Identify whether the cash flow is premium, death benefit, bonus value or withdrawal value.
3. Identify the relevant probability.
4. Identify the relevant discount factor.
5. Write or inspect the Excel formula.
6. Ask Gemini to explain the formula in plain language.
7. Check that the explanation matches the workbook columns.
8. Interpret the result in your own words.

Suggested AI Prompt: Evaluating an Insurance Product with Excel ⋆⋆

Intermediate

"Using Google Gemini as my learning assistant, show me how to evaluate an insurance product using Microsoft Excel formulas. Explain the factors that affect the value of an insurance product and how to measure them. Use an insurance summary table as an example and explain how to calculate the expected PV of premium, face value, face value plus bonus and withdrawal under different scenarios."

Common Mistake / Debugging Warning

Do not ask Gemini only: "What is the answer?" A better student prompt is: "Here is my Excel formula. Please explain whether the cash flow, probability and discount factor are matched correctly." This improves learning and reduces blind copying.

I. Case Study: Designing Products Related to Marital-status Scenarios

Synergy & Learning Flow

Transition to the case study: The first half of Tutorial 11 evaluated an existing life insurance product. The second half asks students to think like product designers. They will use demographic data to design two products related to marital-status scenarios: marriage insurance and single annuity.

Language note for teacher use: We should avoid presenting being single as a problem. The actuarial question is more neutral: different life events and statuses may create different financial needs. The modelling task is to design cash flows that respond to those needs.

Key Definition / Result

Data note: The attached workbook uses the 2011, 2016 and 2021 never-married population proportions included in the Excel file. For this tutorial, treat the workbook as the input dataset. If a newer official table is supplied, update the input sheet first and then recompute the product values.

Source note: Census and Statistics Department. 2021 Population Census. Hong Kong SAR Government. The attached workbook is used as the OSALP input dataset for this case study.

Investigation / Excel Mission: Understand the Product-design Situation

⋆ **Core**

Suppose you are an actuarial trainee in an insurance company. Your team leader asks you to design two products related to financial needs under different marital-status scenarios.

The two products are:

1. **Marriage insurance**
2. **Single annuity**

Before working in Excel, answer the following questions:

1. What event triggers the benefit in marriage insurance?
2. What condition allows the person to keep receiving payments in a single annuity?
3. Why do these two products use the demographic table differently?
4. Why do we need a fair relationship between premium P and benefit c?

Teacher Note: This stage is conceptual. Students should understand the product logic before calculating.

J. Product 1: Marriage Insurance

Synergy & Learning Flow

How this continues Tutorial 10: In Tutorial 10, students learned expected value and expected present value. Marriage insurance applies the same idea to a new event: marriage. The benefit is paid when the marriage event occurs during an age interval.

Key Definition / Result

Marriage insurance structure:

- Pay P to the insurer at ages $19, 24, 29, 34, \dots, 64$.
- Receive c at the time of marriage.
- Aim: to protect against the financial cost triggered by marriage.

In beginner-friendly language, this product says:

You pay regular premiums while exposed to the event,

and

if marriage occurs, the product pays a benefit.

Investigation / Excel Mission: Build the Marriage Insurance Cash-flow Logic

⋆⋆ **Intermediate**

Task. For the marriage insurance product, suggest suitable values of P and c.

Compile the corresponding present values for the insurance. Assume the interest rate equals 3%.

In the solution workbook, one modelling choice is:

$$P = \mathbf{20000}.$$

The workbook then solves for a fair benefit c by comparing the present value of premiums with the expected present value of benefits.

Workbook-based solution values:

$$\text{Marriage Insurance Benefit for Male} \approx \mathbf{128547.05}.$$

$$\text{Marriage Insurance Benefit for Female} \approx \mathbf{110327.13}.$$

Interpretation: The male and female fair benefit values differ because the workbook uses different marriage rates and single rates by sex and age group. The fair benefit is not chosen randomly. It is implied by the probability-weighted and discounted cash-flow structure.

Suggested AI Prompt: Designing Marriage Insurance with Gemini ⋆⋆
Intermediate

"Gemini, I am designing a simple marriage insurance product in Excel. The policyholder pays a premium at ages 19, 24, 29, 34, and so on until age 64, and receives a benefit if marriage occurs. Please explain how to use event probabilities, discount factors and expected present value to choose a fair benefit amount. Use beginner-friendly language and avoid advanced actuarial notation."

Reflection Question: 1

Marriage Insurance Why should the benefit c depend on the probability of marriage at different age intervals?

K. Product 2: Single Annuity

Synergy & Learning Flow

Conceptual contrast: Marriage insurance pays when marriage occurs. A single annuity pays while the person remains single. Therefore, the same demographic table is used differently. One product focuses on the event probability. The other focuses on remaining eligible for payment.

Key Definition / Result

Single annuity structure:

- Pay P to the insurer now.
- Receive c at ages $19, 24, 29, 34, \dots, 64$, until marriage.
- Aim: to protect against the financial need while remaining single.

In beginner-friendly language, this product says:

You make one payment now,

and

you receive regular payments as long as you remain single and eligible.

Investigation / Excel Mission: Build the Single Annuity Cash-flow Logic

⋆⋆ **Intermediate**

Task. For the single annuity product, suggest suitable values of P and c. Compile the corresponding present values for the annuity. Assume the interest rate equals 3%.

In the solution workbook, one modelling choice is:

$$P = \mathbf{5000}.$$

The workbook then solves for a fair periodic payment c.

Workbook-based solution values:

$$\text{Single Annuity Payment for Male} \approx \mathbf{2122.82}.$$

$$\text{Single Annuity Payment for Female} \approx \mathbf{2316.04}.$$

Interpretation: The single annuity values differ because the probability of remaining single differs across age groups and sex in the workbook's input data. The payment c is fair only relative to the assumed premium, probabilities and interest rate.

Suggested AI Prompt: Designing a Single Annuity with Gemini ⋆⋆
Intermediate

"Gemini, I am designing a simple single annuity in Excel. The person pays one premium now and receives periodic payments while still single. Please explain how the probability of remaining single affects the fair annuity payment. Use simple expected present value reasoning and avoid advanced actuarial symbols."

Common Mistake / Debugging Warning

The spelling is **single annuity**, not **single annunity**. Students should use the correct term when writing explanations.

L. Marriage-rate and Single-rate Table

Synergy & Learning Flow

Why this table matters: This table is the product-design version of a life table. Instead of death and survival, we now consider marriage and remaining single. Students should recognise the same modelling structure from the insurance section: event probability and remaining eligible probability.

Key Definition / Result

For this tutorial:

$$q_x = \text{probability of marriage during the age interval.}$$

$$_np_x = \text{probability of still being single at the start of the interval.}$$

A useful plain-English interpretation is:

$$q_x : \text{How likely is the event to happen during this age band?}$$

$$_np_x : \text{How likely is the person still eligible for the next payment or event?}$$

Investigation / Excel Mission: Use the Marriage and Single Status Table
⋆ **Core**

The workbook provides the following table for the product-design exercise.

Age Group	**Male** q_x	**Male** ${}_np_x$	**Female** q_x	**Female** ${}_np_x$
15–19	0.029	1.000	0.047	1.000
20–24	0.104	0.971	0.157	0.953
25–29	0.289	0.867	0.340	0.796
30–34	0.245	0.578	0.204	0.456
35–39	0.105	0.333	0.054	0.252
40–44	0.041	0.228	0.032	0.198
45–49	0.031	0.187	0.009	0.166
50–54	0.037	0.156	0.019	0.157
55–59	0.039	0.119	0.034	0.138
60–64	0.080	0.080	0.104	0.104

Task. Students should answer:

1. Which age group has the largest male marriage rate?
2. Which age group has the largest female marriage rate?
3. How does the probability of still being single change as age increases?
4. Why does the single annuity use ${}_np_x$ more directly than q_x?
5. Why does the marriage insurance use q_x directly when calculating expected benefit?

Teacher Note: Encourage students to describe patterns before calculating product values.

Suggested AI Prompt: Interpreting the Marriage-rate Table ⋆ **Core**

"Gemini, I have a table with age groups, marriage rates and single rates for male and female populations. Please help me interpret the table in plain language. Which columns behave like event probabilities, and which columns behave like eligibility or survival probabilities? Explain how this is similar to a life table but not exactly the same."

M. Worked Example: Male Single Annuity with Interest Rate 0

Synergy & Learning Flow

Why include this simplified example: Before using discounting at 3%, students benefit from a zero-interest example. When the interest rate is 0, there is no discounting. This makes the single annuity logic easier to see.

Worked Example: Male Single Annuity with Interest Rate 0 ⋆ Core

Procedure:

- Pay P to the insurer now.
- Receive \$100 at ages $19, 24, 29, 34, \ldots, 64$, until marriage.

Question: What should be the fair value of P?

If the interest rate is 0, then discounting does not change the value of future payments. The fair premium can be approximated by summing:

$$\sum_{i=1}^{10} 100 \times {}_n p_x.$$

Plain-English explanation: For each possible payment age, the product pays \$100 only if the person is still single and eligible at that age. Therefore, each payment is weighted by the probability of still being single.

Teacher Note: This example should be used to isolate probability weighting before students return to the full 3% discounting model.

Suggested AI Prompt: Understanding the Zero-interest Example ⋆ Core

"Gemini, I am studying a simple single annuity example with interest rate 0. The annuity pays \$100 at several ages if the person is still single. Please explain why the fair premium is the sum of 100 times the probability of still being single at each payment age. Use an intuitive explanation and avoid complicated notation."

N. Full Product-design Task: Compare Marriage Insurance and Single Annuity

Synergy & Learning Flow

Purpose of this integrated task: This is the main actuarial design challenge. Students should use Excel not only to calculate, but to compare product logic, explain fairness and reflect on how assumptions affect product values.

Investigation / Excel Mission: Design Two Products and Compile PVs ⋆⋆⋆

Challenge

Suppose you are an actuarial trainee in an insurance company. Your team leader asks you to design two products related to financial needs under different marital-status scenarios.

Product 1: Marriage Insurance

- Pay P to the insurer at ages $19, 24, 29, 34, \ldots, 64$.
- Receive c at the time of marriage.
- Aim: to protect against the financial cost triggered by marriage.

Product 2: Single Annuity

- Pay P to the insurer now.
- Receive c at ages $19, 24, 29, 34, \ldots, 64$, until marriage.
- Aim: to protect against the financial need while remaining single.

Task. For each of the above two products, suggest suitable P and c. Compile the corresponding present values for the insurance. Assume interest rate equals 3%.

Minimum requirements for student work:

1. Identify input assumptions clearly.
2. Use the marriage-rate and single-rate table.
3. Use a discount factor column.
4. Calculate expected present value row by row.
5. Sum the row-level expected present values.
6. Compare male and female results.

7. Write a plain-English interpretation.
8. Use at least one model check.
9. Use Gemini to explain or debug one formula, then verify the answer manually.

Teacher Note: Do not allow students to submit only final numbers. They must show formula logic, Excel structure and interpretation.

Suggested AI Prompt: Full Product Design Support ⋆⋆⋆ **Challenge**

"Using Google Gemini as my learning assistant, show me how to design an insurance product using Microsoft Excel formulas. Explain the objectives and constraints of designing an insurance product and how to compare the present values of premiums and benefits. Use marriage insurance and single annuity as examples. Suggest how I can choose suitable values for P and c, compare the corresponding PVs and explain why the results are reasonable."

Suggested AI Prompt: Checking Fairness of Premium and Benefit ⋆⋆⋆ **Challenge**

"Gemini, I chose a premium P and calculated a fair benefit c for a marriage insurance product and a single annuity product. Please help me check whether my reasoning is fair. Focus on expected present value, probability weighting, discounting and whether my conclusion depends on the assumptions."

O. Excel Modelling Hints for Students

Synergy & Learning Flow

Why this section matters: Some students may understand the actuarial idea but still make spreadsheet errors. This section gives practical Excel-style guidance using the same modelling spirit as Tutorial 10.

Key Definition / Result

Recommended worksheet structure:

1. **Input section:** interest rate, premium P, benefit c, selected sex.

2. **Data section:** age groups, marriage rates, single rates.
3. **Calculation section:** discount factors, probability-weighted benefits, premium present values.
4. **Output section:** total PV of premium, total EPV of benefit, profit/loss or fair value.
5. **Check section:** probability checks, total checks, missing-value checks.

Investigation / Excel Mission: Suggested Excel Formulas ⋆⋆ Intermediate

Discount factor, if term is in `B2` and interest rate is named `Interest_Rate`:

```
=1/(1+Interest_Rate)^B2
```

Marriage insurance expected benefit, if benefit is named `Benefit`, marriage rate is in `C2`, single rate is in `D2`, and discount factor is in `E2`:

```
=Benefit*C2*D2*E2
```

Single annuity expected payment, if annuity payment is named `Payment`, single rate is in `D2`, and discount factor is in `E2`:

```
=Payment*D2*E2
```

Total expected present value:

```
=SUM(F2:F11)
```

Probability check:

```
=IF(AND(C2>=0,C2<=1,D2>=0,D2<=1),"OK","Check probability")
```

Teacher Note: Column letters may differ depending on students' worksheet layouts. The important issue is not copying these exact addresses, but matching each formula to the correct cash flow, probability and discount factor.

Common Mistake / Debugging Warning

The most common spreadsheet mistakes in this tutorial are:

- mixing up marriage rate and single rate;
- using the wrong sex column;

- copying formulas without fixing assumption cells;
- forgetting to discount future payments;
- using a benefit column instead of a premium column;
- reading a row value as a total value;
- mixing up no-bonus PnL and with-bonus PnL.

Suggested AI Prompt: Debugging Product-design Formulas ⋆⋆ **Intermediate**

"Gemini, here is my Excel formula for an actuarial product-design exercise. Please check whether I have used the correct cash flow, correct probability column and correct discount factor. Also help me identify whether any cell reference should be fixed with dollar signs before using Autofill."

P. Model Checking and Sensitivity Thinking

Synergy & Learning Flow

Connection with Tutorial 10: Tutorial 10 taught that a spreadsheet model should be checkable and changeable. Tutorial 11 now uses that habit in a more realistic actuarial setting.

Investigation / Excel Mission: Build Model Checks ⋆⋆ **Intermediate**

Task. Add at least four checks to your insurance or annuity workbook.

Check 1: Probability between 0 and 1

```
=IF(AND(C2>=0,C2<=1),"OK","Check probability")
```

Check 2: Discount factor positive

```
=IF(E2>0,"OK","Check discount factor")
```

Check 3: Missing age group

```
=IF(A2="","Missing age group","OK")
```

Check 4: Total reconciliation

```
=IF(ABS(F12-SUM(F2:F11))<0.01,"OK","Check total")
```

Teacher Note: Encourage students to deliberately create one error and observe whether their model checks catch the error.

Investigation / Excel Mission: Sensitivity Test ⋆⋆⋆ **Challenge**

Task. Test how your product values change when assumptions change.

1. Change the interest rate from 3% to 2%.
2. Change the interest rate from 3% to 5%.
3. Increase all marriage rates by 10%, but cap them at 100%.
4. Decrease all single rates by 10%, but keep them non-negative.
5. Compare the original and stressed values.

Example formula for increasing a probability by 10%, capped at 100%:

```
=MIN(C2*1.1,1)
```

Example formula for decreasing a probability by 10%, floored at 0%:

```
=MAX(D2*0.9,0)
```

Teacher Note: The goal is not prediction. The goal is to understand that actuarial outputs are sensitive to assumptions.

Suggested AI Prompt: Sensitivity Testing with Gemini ⋆⋆⋆ **Challenge**

"Gemini, I changed the interest rate and probability assumptions in my insurance and annuity Excel model. Please help me explain how these assumption changes affect the expected present value. Remind me that sensitivity testing is not the same as predicting the future."

Q. Written Interpretation Practice

Synergy & Learning Flow

Why this section matters: Actuaries do not only calculate. They explain. Students should practise writing short professional interpretations of spreadsheet outputs.

Investigation / Excel Mission: Write an Interpretation for the Insurance Evaluation ⋆⋆ **Intermediate**

Task. Write a short paragraph explaining the no-withdrawal insurance result.

Your paragraph should mention:

1. expected PV of premium;
2. expected PV of face value;
3. expected PV of face value plus bonus;
4. why the values differ;
5. why probability and discounting are both needed.

Suggested answer structure:

> Under the no-withdrawal scenario, the expected PV of premiums is approximately ..., while the expected PV of the face value is approximately .... When bonus is included, the expected PV of the benefit becomes much larger. This happens because the bonus-adjusted benefit can grow over time. However, all future cash flows must still be weighted by probabilities and discounted back to today's value.

Investigation / Excel Mission: Write an Interpretation for the Product Design Task ⋆⋆ **Intermediate**

Task. Write a short paragraph comparing marriage insurance and single annuity.

Your paragraph should mention:

1. how marriage insurance pays when the event occurs;
2. how single annuity pays while the person remains eligible;
3. why q_x and ${}_np_x$ are used differently;
4. why male and female values may differ in the workbook;
5. why the selected P and c should be interpreted under the assumptions.

Suggested AI Prompt: Improving My Interpretation ⋆⋆ **Intermediate**

"Gemini, I wrote a paragraph interpreting my insurance and annuity Excel results. Please help me make the explanation clearer for a secondary school audience. Do

not change my meaning. Make sure I explain probability, discounting, premium, benefit and assumptions clearly."

R. Consolidation and Exit Questions

Investigation / Excel Mission: Insurance Evaluation Review ⋆ **Core**

1. What is face value?
2. What is policy bonus?
3. What is withdrawal value?
4. Why do we need a death rate in a life insurance model?
5. Why do we need a survival rate?
6. Why do we discount future cash flows?
7. What three components are multiplied in expected present value?
8. What is the difference between a row-level EPV and a cumulative total EPV?

Investigation / Excel Mission: Withdrawal Decision Review ⋆⋆ **Intermediate**

1. What changes in the cash-flow model when the insured withdraws?
2. Why should later death benefits not be counted after withdrawal?
3. What is the best no-bonus withdrawal timing in the workbook?
4. What is the worst no-bonus withdrawal timing in the workbook?
5. What is the best with-bonus withdrawal timing in the workbook?
6. What is the corrected worst with-bonus withdrawal value in the workbook?
7. Why should students be careful not to mix up PnL columns?

Investigation / Excel Mission: Marriage Insurance and Single Annuity Review ⋆⋆ **Intermediate**

1. What event triggers the marriage insurance benefit?
2. What condition allows the single annuity payment to continue?
3. What does q_x mean in this tutorial?
4. What does ${}_np_x$ mean in this tutorial?
5. Why does marriage insurance use event probability?
6. Why does single annuity use remaining-single probability?
7. Why should premium and benefit be compared using expected present value?

Investigation / Excel Mission: Excel Modelling Review ⋆⋆ **Intermediate**

1. Why should assumptions be placed in visible input cells?
2. Why should some references be fixed with dollar signs or Named Ranges?
3. Why is Autofill useful but also risky?
4. What model checks did you add?
5. What sensitivity test did you perform?
6. What changed when the interest rate changed?
7. What changed when the probability assumptions changed?

Investigation / Excel Mission: AI-supported Learning Review ⋆⋆ **Intermediate**

1. How did Gemini help you understand a formula?
2. How did Gemini help you debug a spreadsheet issue?
3. Why should you not blindly trust an AI-generated formula?
4. What evidence did you use to verify the AI response?
5. Write one AI prompt that helped you learn rather than copy.

Investigation / Excel Mission: Challenge Questions ⋆⋆⋆ **Challenge**

1. If the interest rate increases, what usually happens to the present value of future benefits? Why?
2. If marriage rates increase, how might the fair marriage insurance benefit change?
3. If single rates decrease, how might the fair single annuity payment change?
4. Can a product be fair under one assumption set but unfair under another assumption set?
5. Why is actuarial modelling not just mathematics but also judgement?
6. How would you explain expected present value to a younger student?
7. What makes this spreadsheet actuarial in spirit?

S. Final Takeaway

Key Definition / Result

Final Takeaway. Tutorial 11 continues the spreadsheet-modelling foundation from Tutorial 10 and applies it to actuarial insurance and annuity valuation.

The most important actuarial pattern is:

$$\text{Expected Present Value} = \text{Cash Flow} \times \text{Probability} \times \text{Discount Factor}.$$

The most important spreadsheet habit is:

Do not only calculate. Build a model that can be checked, changed and explained.

The most important AI-learning habit is:

Do not ask AI only for answers. Ask AI to explain, challenge and debug your reasoning.

By the end of this tutorial, students should be able to read an actuarial-style workbook, explain insurance variables, calculate expected present values, compare withdrawal decisions, design simple marriage insurance and single annuity products, run basic model checks, perform sensitivity tests and write clear interpretations of their results.

End of Tutorial 11